

2026-05-30 - KXTJ3-1057 Accelerometer with USB Shell

Goal

Implement a custom I2C driver for the KXTJ3-1057 3-axis accelerometer and expose it through an interactive Zephyr Shell over USB CDC-ACM.

What you will learn

- How to write a minimal custom I2C driver when no Zephyr driver exists
- How to read raw accelerometer data and convert it to g-force
- How to combine USB CDC-ACM and the Zephyr Shell in one application
- How to build shell commands that read and configure hardware

KXTJ3-1057 Overview

The KXTJ3-1057 (Kionix) is a low-power 3-axis MEMS accelerometer: - Interface: I2C (up to 400 kHz) - I2C address: **0x0E** (ADDR/SDO pin low) or **0x0F** (ADDR/SDO pin high) - Measurement ranges: $\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$ - Output: 12-bit or 8-bit resolution - WHO_AM_I register: **0x0F** $\rightarrow$ value **0x35**

Key registers

Register	Address	Description
XOUT_L	0x06	X-axis low byte
XOUT_H	0x07	X-axis high byte
YOUT_L	0x08	Y-axis low byte
YOUT_H	0x09	Y-axis high byte
ZOUT_L	0x0A	Z-axis low byte
ZOUT_H	0x0B	Z-axis high byte
WHO_AM_I	0x0F	Device ID (0x35)
CTRL_REG1	0x1B	Power/range/resolution control

CTRL_REG1 bit map

Bit 7: PC1 – 1 = operating mode, 0 = standby
Bit 6: RES – 1 = 12-bit, 0 = 8-bit
Bit 4: GSEL1 – range select MSB

Bit 3: GSEL0 – range select LSB

00 = $\pm 2g$, 01 = $\pm 4g$, 10 = $\pm 8g$ (14-bit), 11 = $\pm 8g$

Hardware wiring (nRF52840 DK)

KXTJ3-1057 pin nRF52840 DK pin

VDD	3.3V
GND	GND
SDA	P0.26
SCL	P0.27
ADDR	GND ($\rightarrow 0x0E$)

Project structure

2026-05-30_KXTJ3_Accel_USB_Shell/

├── CMakeLists.txt	
├── prj.conf	
├── app.overlay	
├── drivers/	
│ ├── kxtj3/	
│ │ ├── kxtj3.h	← register map and API
│ │ └── kxtj3.c	← I2C read/write implementation
└── src/	
└── main.c	← USB init + shell commands

prj.conf

```
CONFIG_I2C=y
CONFIG_USB_DEVICE_STACK=y
CONFIG_USB_CDC_ACM=y
CONFIG_UART_LINE_CTRL=y
CONFIG_SHELL=y
CONFIG_SHELL_BACKEND_SERIAL=y
CONFIG_LOG=y
CONFIG_LOG_DEFAULT_LEVEL=3
```

Shell commands implemented

Command	Description
accel whoami	Read WHO_AM_I register (expect 0x35)
accel read	Print X, Y, Z in mg
accel stream <n>	Print n readings at 200 ms intervals
accel range <2\ 4\ 8\ 16>	Set measurement range

Converting raw counts to mg

The 12-bit result is left-justified in a 16-bit register pair. Shift right by 4 to get the signed 12-bit value, then scale:

```
int16_t raw = (int16_t)((high << 8) | low) >> 4;
int32_t mg   = (raw * 1000) / counts_per_g;
// counts_per_g = 1024 for ±2g, 512 for ±4g, 256 for ±8g
```

Why this matters

Many sensors used in production products are not supported by Zephyr's upstream drivers. Writing a thin I2C wrapper is a core embedded skill. The shell interface lets you validate sensor output during bring-up without rebuilding firmware.

Practice tasks

1. Verify WHO_AM_I returns 0x35 before reading data.
2. Hold the board still and confirm Z reads close to 1000 mg (1g).
3. Try `accel range 4` and confirm the readings change scale.
4. Extend `accel stream` to also encode data as CBOR (combine with Day 12).

Example folder

See `src/2026-05-30_KXTJ3_Accel_USB_Shell` for the full driver and application code.

Next topic

Future topics: BLE peripheral advertising, MCUboot firmware updates, low-power sleep modes.